

GOKUL R.

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ABOUT ME

Third-year Artificial Intelligence and Machine Learning undergraduate with hands-on experience across computer vision, reinforcement learning, and full-stack development. Built a hybrid YOLOv8 + R-CNN pipeline for real-time medical video analysis, a PPO reinforcement-learning controller for robotic manipulation, and deployed applications using FastAPI, Docker, and PostgreSQL. Seeking an AI/ML Engineer, Computer Vision, or Applied AI internship role.

PROJECTS

AI-Assisted Colonoscopy Polyp Detection — Hybrid Deep Learning

2026 – Present

YOLOv8, R-CNN, PyTorch, OpenCV, Temporal Consistency Filtering

- Built a hybrid YOLOv8 + R-CNN pipeline for real-time polyp detection in colonoscopy video, targeting 40+ FPS on edge hardware.
- Engineered temporal consistency filtering across video frames to suppress false positives and stabilize bounding-box predictions in high-motion endoscopic sequences.
- Benchmarking detection latency against precision-recall trade-offs for integration with clinical endoscopy workflows.

Vision-Guided Reinforcement Learning for 7-DOF Robotic Arm

2026

PyTorch, PyBullet, PPO, ResNet-18

- Built a perception-to-action pipeline combining a ResNet-18 visual backbone with PPO actor-critic control for a 7-DOF Franka Panda simulation.
- Applied visual and dynamics domain randomization (lighting, sensor noise, damping, friction) for sim-to-real transfer; reached 92% success at 650mm target error and ~81 FPS inference.

RTS Wedding — Full-Stack Wedding Invitation Website

2026

HTML5, CSS3, JavaScript, Netlify

- Built and deployed a responsive, animated wedding invitation web app with RSVP functionality, dynamic countdowns, and smooth scroll animations; hosted on Netlify.

ChakraPulse — Distributed Telemetry Backend

2025 – 2026

PostgreSQL, Docker, Python, FastAPI

- Engineered a containerized telemetry backend system for collecting and processing sensor data streams with PostgreSQL storage and Docker deployment.

SKILLS

Programming & Web: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, React, Next.js, Node.js

AI / ML Frameworks: PyTorch, TensorFlow, YOLOv8, R-CNN, Deep Learning, Reinforcement Learning (PPO), Computer Vision, NLP, Transformers

Tools & Cloud: Git/GitHub, Docker, CI/CD, FastAPI, PostgreSQL

Hardware & Embedded: Arduino, Raspberry Pi, Microcontrollers, Servo Actuation, Sensor Interfacing, IoT

CERTIFICATIONS

- NPTEL Elite + Gold, Human-Computer Interaction
- Cisco Network Addressing & AI Fundamentals

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning

2024 – 2028

Easwari Engineering College (SRM Group), Chennai, India — CGPA: 7.9 / 10

EXPERIENCE & LEADERSHIP

IoT & Sensors Intern — PrimeOxy

Selected (Awaiting Offer)

- Selected for embedded sensor interfacing and IoT data pipelines as part of a structured internship track.

President — Leo Club, SRM Easwari Engineering College

Jun 2025 – Mar 2026

- Led a student volunteer body coordinating community service and campus outreach initiatives, managing a team and event budgets.

ACHIEVEMENTS

- **Top 5 of 300+ teams, TRIGENIX-26 National Hackathon (March 2026)** – autonomous AI solution, 24-hour non-stop hackathon.
- **Grand Finals Qualifier, Triwizardathon 1.0 (August 2025)** – robotic-arm solution with servo actuation and microcontroller control, built in 24 hours. Microsoft Learn Student Ambassador event.
- **Chemstar Talent Exam – 2nd Prize (March 2025)** – Dept. of Chemistry, Easwari Engineering College.